# Tan Dai Ngo

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## EDUCATION

**University of Victoria** | MEng, Applied Data Science (Co-op) *Victoria, BC | Sep 2025 - Dec 2027*

- **Coursework:** Optimization for Machine Learning, Algorithms and Data Models, Data Mining. **GPA:** 8.78 / 9.00.

**University of Chicago** | MS, Computer Science *Chicago, IL, USA | Sep 2020 - Mar 2022*

- **Coursework:** Algorithms, Distributed Systems, Databases, Discrete Mathematics, Applied Software Engineering.

**University of Washington** | BS, Economics *Seattle, WA, USA | Sep 2015 - Aug 2019*

- **Minors:** Informatics, Applied Mathematics. **Certificate:** Quantitative Managerial Economics. **GPA:** 3.57 / 4.00

## EXPERIENCE

**Software Developer** | T-Mobile (via BeaconFire Inc.) *Bellevue, WA, USA | Jun 2022 - Apr 2024*

- Developed and maintained REST endpoints across 5+ Roaming Business System microservices in Java and Spring Boot.
- Automated reprocessing of rejected roaming tests with weekly Kafka retries, cutting manual effort by 80%.
- Traced 20+ cross service defects to their root cause and deployed features via Jenkins across 50+ releases.
- Tuned Cassandra queries across multiple data sources to respond in under 1 second, debugging in Splunk.

**Math and Computer Science Teacher** | APU American International Schools *Da Nang, Vietnam | Aug - Dec 2024*

- Explained technical material to 50+ students with no prior background, adjusting the pitch in real time per student.
- Mentored students one-on-one on debugging and algorithmic reasoning by working through their own broken code.

## PROJECTS

**Multi-Stage IoT Intrusion Detection** | Python, XGBoost, scikit-learn *UVic Capstone, team of 7 | May - Aug 2026*

- Compared six supervised classifiers on 206,000 CIC IoT-DIAD 2024 flows with preprocessing fit on training only.
- Selected binary XGBoost under a 1% false positive budget, reaching 99.5% recall at 0.94% FPR and 0.989 PR-AUC.
- Designed a four condition holdout that exposed capture session leakage, dropping PR-AUC from 0.981 to 0.125.

**Ocean Data ML Platform with RAG** | Python, Spark, XGBoost, Qdrant, FastAPI *Personal Project | Feb - Mar 2026*

- Harmonized 10.8 million NOAA and Ocean Networks Canada records into Bronze, Silver and Gold layers with lineage.
- Diagnosed the data as the limit on water temperature while the same pipeline cut air temperature RMSE by 48%.
- Served semantic search behind FastAPI, Qdrant and Ollama at hit@k 0.9 and term recall 0.85 on held out queries.

**Auction Marketplace Microservices** | Python, Flask, MySQL, MongoDB, RabbitMQ *UChicago, team of 4 | Feb - Mar 2021*

- Split a marketplace into six Flask services with isolated MySQL schemas, preventing changes from breaking callers.
- Routed durable messages through RabbitMQ and stored them in MongoDB across six independently deployed services.
- Containerized the full stack with 14 Docker containers, dependency-ordered startup, and a shared bridge network.

**Distributed Facility Reservation System** | Python, FastAPI, SQLite, pytest *UChicago, team of 4 | Apr - May 2021*

- Delivered 27 versioned REST endpoints with OpenAPI docs and SQLite booking rules for conflicts and cancellations.
- Validated API, database, and booking logic with 71 pytest tests rerun on every code change across all three layers.
- Negotiated one shared booking contract with four teams, then tested against the services they built independently.

## TECHNICAL SKILLS

**Programming Languages:** Python, Java, SQL, Cypher, MATLAB, JavaScript, HTML, CSS
**Machine Learning & Evaluation:** PyTorch, XGBoost, CatBoost, scikit-learn, NumPy, Optuna, MLflow, Sentence Transformers
**Data, Retrieval & Messaging:** Spark, PySpark, ETL, Qdrant, RAG, embeddings, MySQL, MongoDB, Cassandra, RabbitMQ, Kafka
**Backend, Cloud, CI/CD & Testing:** Flask, FastAPI, Spring Boot, REST, OpenAPI, Docker, AWS, SQLite, Git, Jenkins, pytest
**Credentials:** Google Advanced Data Analytics (2025), Shell.ai Hackathon 2025, IMC 64 Bids (top 250 of 1,500)